

Arakawa–Schubert (1974): goal of the paper

1

Large-scale models need the combined heating and moistening effects of unresolved cumulus clouds.

Goal Compute these cloud effects from the large-scale atmosphere itself.

How: three steps

slide 2

Step 1: budgets

The environmental budgets reduce the cloud effects to three vertical profiles.



slides 3–4

Step 2: cloud types

Compute each cloud type, then combine them with one mass-flux spectrum.



slides 5–7

Step 3: closure

A buoyancy balance determines the spectrum. Tropical data test it.

unknowns left

3 profiles: $M_c(z)$, $D(z)$, $\hat{l}(z)$

unknowns left

1 spectrum: $\mathcal{M}_B(\lambda)$

unknowns left

none: the system is closed

The environment sets the cloud activity, which in turn changes the environment

Step 1: clouds enter through three profiles

2

Budgets of the air between clouds: the model knows every term except M_c , D and $\hat{l}$.

$$\rho \partial_t \bar{s} = D(\hat{s} - \bar{s} - L\hat{l}) + M_c \partial_z \bar{s} + \text{LS}_s$$

$$\rho \partial_t \bar{q} = D(\hat{q}^* + \hat{l} - \bar{q}) + M_c \partial_z \bar{q} + \text{LS}_q$$

Symbols

$s = c_p T + gz$, q dry static energy; vapour mixing ratio

$\bar{s}$, $\bar{q}$ large-scale means, i.e. the environment

$M_c(z)$ total upward cloud mass flux per unit area

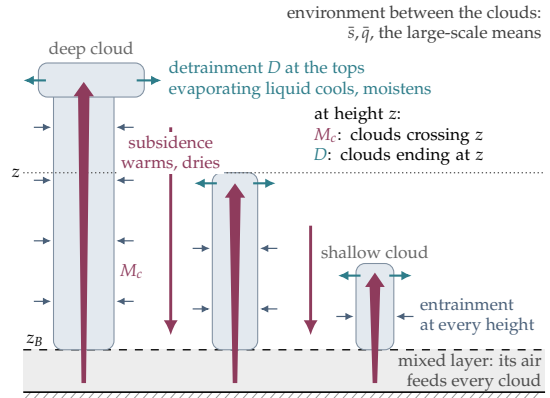
$D(z)$ mass detrained per unit volume and time

$\hat{s}$, $\hat{q}^*$ detrained-air values (star: saturation)

$\hat{l}(z)$ liquid water in the detrained air

L , ρ latent heat of vaporization; air density

LS large-scale advection (+ radiation for s)



Unknowns left: three profiles, $M_c(z)$, $D(z)$ and $\hat{l}(z)$

Step 2: the structure of one cloud type

3

Entrainment dilutes a rising plume. The more it entrains, the lower its top.

$$\partial_z \eta = \lambda \eta$$

mass grows: $\eta = e^{\lambda(z-z_B)}$

$$\partial_z h_c = \lambda(\bar{h} - h_c)$$

h_c is pulled toward $\bar{h}$ over a length $1/\lambda$

$$h_c(z_D) = \bar{h}^*(z_D)$$

top $z_D(\lambda)$: buoyancy vanishes

All types start at z_B with mixed-layer air: $\eta = 1, h_c = h_M$.

Symbols

λ fractional entrainment rate: the cloud-type label

η mass flux normalized at the mixed-layer top z_B

$h = s + Lq$ moist static energy; h_M : its mixed-layer value

$\bar{h}, \bar{h}^*$ environment h ; the h it would have if saturated

$h_c > \bar{h}^*$ plume warmer than the environment, so buoyant

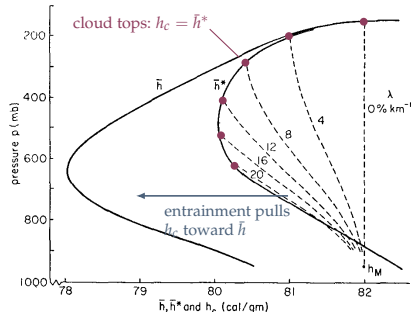


Fig. 6, tropical sounding; up is higher. Dashed: plume h_c for $\lambda = 0\text{--}20\% \text{ km}^{-1}$, all from h_M . Dots (added): cloud tops.

Given the environment and λ , the plume model fixes η , h_c and the cloud top z_D

Step 2: one spectrum gives the three profiles

4

The plume model gives each type's structure; the spectrum $\mathcal{M}_B(\lambda)$ weights the types.

Clouds still rising at height z

$$M_c(z) = \int_0^{\lambda_D(z)} \mathcal{M}_B(\lambda) \eta(z, \lambda) d\lambda$$

Clouds ending at height z

$$D(z) = -\mathcal{M}_B(\lambda_D) \eta(z, \lambda_D) \frac{d\lambda_D}{dz}$$

$$\hat{l}(z) = l(z, \lambda_D)$$

Symbols

$\mathcal{M}_B(\lambda) d\lambda$ mass flux at z_B of the types in $(\lambda, \lambda + d\lambda)$

$\eta(z, \lambda)$ normalized mass flux (slide 3)

$\lambda_D(z)$ type whose cloud top is at z

$l(z, \lambda)$ liquid water inside type λ (below)

Where l comes from

Plume water budget (Eq. 85): vapour in by entrainment, rain r out.

$$\partial_z [\eta (q_c + l)] = \lambda \eta \bar{q} - \eta r$$

Cloud vapour q_c is saturated (Eq. 90), so the rest is l .

Only the mass-flux spectrum $\mathcal{M}_B(\lambda)$ remains to be determined

Step 3: the cloud work function

5

The work function $A(\lambda)$ adds up buoyancy over the cloud depth, weighted by mass flux. For each cloud type, the environment determines A .

$$A(\lambda) = \int_{z_B}^{z_D(\lambda)} \underbrace{\eta(z, \lambda)}_{\text{weight}} \underbrace{\frac{g}{c_p \bar{T}(z)} [s_{vc}(z, \lambda) - \bar{s}_v(z)]}_{\text{buoyancy}} dz$$

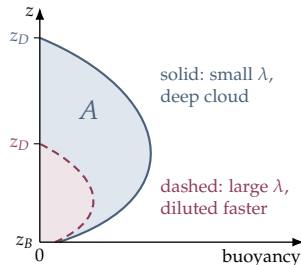
Symbols

$s_v = s + c_p \bar{T} (0.608 q - l)$ virtual static energy (Eq. 58)

T, q, l temperature, vapour, liquid water; c : cloud, bar: environment

g, c_p gravity; specific heat of air at constant pressure

η mass flux normalized at the mixed-layer top z_B



Schematic. A : area under the buoyancy curve, weighted by η .

Why “work”: kinetic energy

$$dK/dt = A \mathcal{M}_B - \mathcal{D}$$

Positive A allows buoyant energy production, but does not set the cloud activity

Step 3: the quasi-equilibrium closure

6

The rate of change of A is the sum of the large-scale forcing and the cloud response.

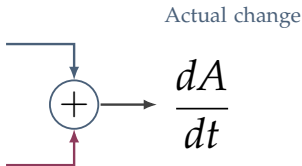
$$\frac{dA(\lambda)}{dt} = \int_0^{\lambda_{\max}} K(\lambda, \lambda') \mathcal{M}_B(\lambda') d\lambda' + F(\lambda)$$

Forcing $F(\lambda)$

Large-scale processes

Cloud response

All types λ' , through K



Kernel $K(\lambda, \lambda')$

Change of $A(\lambda)$ per unit time due to unit cloud-base mass flux of type λ' , acting through the environment. Usually $K < 0$: clouds use up A .

K_V mostly < 0 : subsidence driven by λ' warms the air, so λ is less buoyant

$K_D \geq 0$: shallower clouds ($\lambda' > \lambda$) detrain, cooling and moistening the air of deeper ones

K_M mass drawn up at z_B changes the mixed-layer depth

Quasi-equilibrium: cloud and forcing terms nearly cancel, giving $dA/dt \simeq 0$

dA/dt : change of A per unit time. Observed residual: $|dA/dt| \ll |F_C|$.

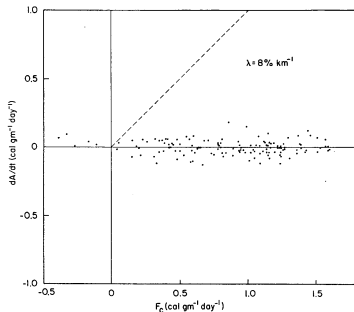


Fig. 13. F_C : cloud-layer forcing. Dashed line: $dA/dt = F_C$ without cloud response.

τ_{adj} : cloud adjustment time

Defined in words (§7): time for the clouds alone, with no forcing, to bring A to neutral ($A = 0$). Estimated from typical A and cloud term (Eqs. 152–154): $\tau_{\text{adj}} \sim 10^3\text{--}10^4$ s, 15 min–3 h.

τ_{LS} : forcing time

Defined in words: time over which F changes significantly. Stated as typical, not derived: $\tau_{\text{LS}} \gtrsim 10^5$ s, a day or more.

Why the net change is small (Eqs. 146–148)

cloud term	$ [dA/dt]_C \sim A/\tau_{\text{adj}}$	clouds could remove A in τ_{adj}
net change	$ dA/dt \lesssim A/\tau_{\text{LS}}$	A follows the slow forcing
ratio	$\lesssim \tau_{\text{adj}}/\tau_{\text{LS}} \ll 1$	cloud term and F nearly cancel

$\tau_{\text{adj}} \ll \tau_{\text{LS}}$: clouds remove A far faster than A changes, so they nearly cancel F